

UNIVERSAL INDUCTION TROUBLESHOOTING FRAMEWORK

CORE FAULT LOGIC FOR AI DIAGNOSTIC ENGINE INTEGRATION

Module 1: Ground Fault & Leakage Diagnostics

Core Principle: A system shutdown is triggered when the ground leakage current surpasses the internal reference trip level.

Baseline Metrics: A standard furnace system completely drained of cooling water will typically show a resistance to ground of approximately 40 k Ohms. When the system is operational and filled with cooling water, this resistance normalizes between 5 k Ohms and 10 k Ohms, dictated by the specific conductivity of the water.

Diagnostic Path:

- **Refractory Penetration (Ghost Faults):** If the system exhibits high resistance to ground while powered down but immediately trips upon energizing, suspect thermal expansion or a spark-over event caused by metal penetrating the lining.
- **Yoke Isolation:** To isolate a defective yoke, systematically loosen the yoke bolts one at a time while continuously monitoring the resistance readings.
- **Physical Contamination:** Thoroughly inspect the physical architecture for conductive scrap lodged between the furnace frame and the coil conductor, and verify there are no water leaks at the power cable flanges.
- **Capacitor Breakdown:** If the fault is not located at the furnace, isolate the power supply and disconnect the capacitor bank one group at a time to identify any shorted or leaking capacitors.
- **Monitor Circuitry:** Verify the integrity of the ground leakage monitor choke by checking the resistance of its leads to ground, and test the suppression diodes.

Module 2: Overvoltage & Inverter Trips

Core Principle: The control logic aborts the run state when it detects that the peak furnace voltage has exceeded the normal maximum operating threshold.

Diagnostic Path:

- **Semiconductor Failure:** Inspect all inverter Silicon-Controlled Rectifiers (SCRs) and replace any that are found defective. Note that pressure-mounted SCRs must be tested under their normal clamping pressure, as a shorted unit may falsely test as functional if removed from the assembly.
- **Regulation Instability:** Investigate the system's voltage regulation parameters, as instability here can trigger overvoltage limits.
- **Transformer Degradation:** Test the inverter potential transformer and replace it if it is outputting incorrect reference voltages.
- **Control Processor Malfunction:** If hardware tests clear, swap the main digital control card with a pre-programmed spare to rule out corrupted logic or EPROM failure.

Module 3: Overcurrent & System Overloading

Core Principle: A hard fault is triggered if the peak value of the input current surges more than 25% above the established maximum operating capacity.

Diagnostic Path:

- **Rectifier SCR Degradation:** Test every rectifier SCR for short circuits. Ensure the gate lead to anode resistance measures at 15 k Ohms or higher.
- **Snubber Circuitry:** Inspect the AC/DC snubber circuits, specifically testing for defective capacitors or shorted diodes.
- **Synchronization Supply:** Check the synchronizing transformer for proper voltage output and verify the integrity of its fuses.

- **Pulse Transformer Logic:** Verify that all pulse transformers are connected correctly. Check the primary side for approximately 48 ohms of resistance and verify the secondary side reads open or as a diode in one direction.